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Under review as a conference paper at ICAIS 2025

# REVOLUTIONIZING AI CONFERENCE PEER REVIEW: A BI-DIRECTIONAL FEEDBACK AND REWARDS FRAMEWORK

**Anonymous authors**  
Paper under double-blind review

## ABSTRACT

The rapid increase in submissions to AI conferences has led to a crisis in the peer review process, characterized by declining review quality and accountability. This position paper proposes a novel bi-directional feedback mechanism where authors can evaluate the quality of reviews while safeguarding against retaliation. Coupled with a blockchain-enabled reviewer rewards system, this framework aims to incentivize high-quality reviewing and create an accountability structure that benefits all stakeholders. By allowing authors to provide feedback on reviews and rewarding reviewers with transparent digital credentials, this system fosters a culture of quality and responsibility in the peer review process. We call upon the AI community to engage in this vital conversation and explore these transformative reforms for sustainable peer review practices.

## 1 INTRODUCTION

The peer review process is critical for maintaining the integrity of academic publishing, yet it faces increasing challenges due to high submission rates, reviewer burnout, and lack of accountability (Kim et al., 2025; Mishra, 2025). Traditional methods often overlook the need for reciprocal feedback, limiting opportunities for improvement. This paper introduces a bi-directional feedback system that allows authors to evaluate reviews while simultaneously rewarding reviewers through a blockchain mechanism. This dual approach aims to enhance review quality, encourage accountability, and facilitate a more sustainable peer review ecosystem.

## 2 RELATED WORK

Existing literature highlights systemic issues in the peer review process, such as reviewer workload and biases (Mishra, 2025). Recent proposals have addressed accountability and transparency using technology, yet few have integrated a comprehensive feedback loop (V, 2025; Shrestha et al., 2024). This work distinguishes itself by proposing a two-stage feedback loop that empowers authors and incentivizes reviewers, ensured by blockchain technology (Wardani et al., 2025).

## 3 METHOD

The proposed system integrates a bi-directional feedback mechanism wherein authors can anonymously rate the quality of reviews, fostering a culture of accountability. Reviewers receive in the form of blockchain-verified digital credentials based on their performance, enhancing their motivation to provide quality feedback. This model not only incentivizes high-quality reviews but also promotes a transparent system where contributions are recognized and valued.

## 4 EXPERIMENTAL SETUP

To evaluate the effectiveness of the bi-directional feedback system, we designed a pilot study involving a select group of AI conferences. Authors will be able to rate reviews anonymously, and a

1

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blockchain-based digital badge system will track reviewer contributions over time, yielding a quantifiable 'Reviewer Impact Score'. Surveys will be conducted pre- and post-implementation to assess perceived review quality and satisfaction.

## 5 EXPERIMENTS

### Version Management

Version 1

### Citation Tools

Cite This Paper

 **Meta Review Status:** The meta review and decision have not been released yet.

## AI Review from DeepReviewer

 AI Review from DeepReviewer will be automatically processed

### Summary

This paper introduces a novel framework for enhancing the peer review process in AI conferences through a bi-directional feedback mechanism and a blockchain-based reviewer reward system. The core idea is to empower authors to evaluate the quality of reviews they receive, thereby increasing reviewer accountability, and to incentivize high-quality reviewing by awarding reviewers with blockchain-verified digital credentials. The proposed system aims to address the growing challenges of review quality and reviewer burnout in academic publishing. The authors describe a system where authors can rate reviews, and these ratings contribute to a Review Quality Score (RQS) which is then used to determine the rewards reviewers receive. The reward system utilizes blockchain technology to issue digital badges or tokens to reviewers who consistently provide high-quality reviews. The authors conducted experiments to evaluate the effectiveness of their proposed system. These experiments involved training a neural network to predict a synthetic RQS and tuning hyperparameters to optimize the predictive performance. The results of these experiments demonstrated high training and validation accuracy for the baseline model and significant improvements in RQS metrics during hyperparameter tuning. The authors claim that their framework has the potential to improve the overall quality and integrity of the peer review process in academic publishing. However, the paper lacks crucial details regarding the implementation of the proposed system, and the experiments do not directly validate the effectiveness of the bi-directional feedback and reward system in a real-world scenario. The paper's primary contribution lies in its conceptual framework for improving peer review, but its practical implementation and validation require further development.

#### Version Management

Version 1

### Strengths

The paper's primary strength lies in its identification of a critical problem in the current academic publishing landscape: the declining quality of reviews amidst increased submissions and reviewer burnout. The proposed bi-directional feedback mechanism, which allows authors to evaluate reviews, is a novel approach to increasing reviewer accountability. This idea directly addresses the asymmetry in the current peer review system, where reviewers are not evaluated on the quality of their reviews. The introduction of a blockchain-based reward system is also a novel contribution, aiming to incentivize high-quality feedback. The use of blockchain technology for verifying digital credentials is a promising approach to ensure the integrity and transparency of the reward system. The paper's attempt to quantify review quality through a synthetic Review Quality Score (RQS) is a step towards objective evaluation of reviews. The authors also attempted to validate their framework through experiments, which is a positive step towards demonstrating the potential of their approach. The idea of using digital badges or tokens as rewards for reviewers is an innovative way to recognize and appreciate their contributions. The paper's focus on improving the peer review process is highly relevant to the AI community, and the proposed framework has the potential to address some of the critical challenges facing academic publishing.

#### Citation Tools

### Weaknesses

After a thorough review, I've identified several significant weaknesses in this paper that undermine its overall effectiveness. Firstly, the paper lacks clarity and coherence, making it difficult to fully grasp the proposed contribution. The description of the bi-directional feedback mechanism and the blockchain-based reward system feels disjointed, and the connection between these two components is not clearly articulated beyond a high-level description in the 'Workflow' section. The paper reads more like a brainstorming session than a polished academic paper, lacking the necessary detail and structured explanation expected in a scholarly work. The methodology for implementing the feedback mechanism is vague, with no specific details on the rating scale or the platform used. Similarly, the blockchain-based reward system is described in broad terms, without specifying the smart contract logic, the specific blockchain platform, or the process of awarding digital credentials. This lack of detail makes it difficult to assess the feasibility and practicality of the proposed system. My confidence in this assessment is high, as the lack

of detail is evident throughout the 'Method' section. Secondly, the paper fails to provide a detailed explanation of how the blockchain-based reward system would function, the metrics for review quality, and how to prevent potential abuse in the feedback process. The paper mentions that reviewers receive rewards in the form of blockchain-verified digital credentials based on their performance, but it doesn't specify the criteria for earning these credentials beyond 'consistently high-quality reviews.' The specific metrics used to determine review quality are not defined, and the process by which authors rate reviews is not explained. The paper also mentions 'safeguarding against retaliation' but does not elaborate on the mechanisms in place to prevent authors from giving unfairly low ratings or reviewers from manipulating the system. This lack of detail raises serious concerns about the fairness and robustness of the proposed system. My confidence in this assessment is high, as the paper lacks any concrete details on these critical aspects. Thirdly, the experimental setup and results focus on a predictive model for a synthetic 'Review Quality Score' (RQS) rather than evaluating the proposed peer review system itself. The experiments involve training a neural network to predict this synthetic RQS and tuning hyperparameters to optimize predictive performance. However, there is no experiment that directly measures the impact of the feedback and reward system on actual review quality or reviewer behavior. The connection between the experiments and the proposed framework is unclear, and the paper does not provide evidence that the system improves review quality or addresses the challenges mentioned. The experiments do not validate the core claims of the paper, and the use of a synthetic RQS creates a disconnect between the proposed system and the experimental evaluation. My confidence in this assessment is high, as the experiments are clearly not designed to test the effectiveness of the proposed peer review system. Finally, the paper's methodology and implementation plan for the proposed system is vague. There are no case studies, real-world data, or pilot studies to support the feasibility and effectiveness of the framework. The paper describes a 'pilot study involving a select group of AI conferences,' but the details of this pilot study are minimal, with no information on the number of participants, the specific conferences involved, or the data collected. The lack of real-world data or pilot study results makes it difficult to assess the practicality and impact of the proposed framework. The paper also lacks details on the blockchain implementation, including the type of blockchain used, how transactions are recorded, and how privacy is maintained. This lack of specificity makes it difficult to understand how the proposed system would be implemented in practice. My confidence in this assessment is high, as the paper lacks the necessary details to support the feasibility and effectiveness of the proposed system.

 **Suggestions**

Based on the identified weaknesses, I recommend several concrete improvements to strengthen this paper. Firstly, the authors should significantly enhance the clarity and coherence of the paper. This can be achieved by providing a more structured and detailed explanation of the proposed system, including a clear articulation of how the bi-directional feedback mechanism and the blockchain-based reward system interact. The paper should be revised to ensure the connection between the different components is clear and that the overall contribution is easily understandable. The authors should also provide a detailed description of the methodology, including the specific rating scale for reviews, the platform used for feedback, and the criteria for earning rewards. Secondly, the authors need to address the issue of potential abuse in the feedback process. This can be achieved by implementing mechanisms to prevent authors from giving unfairly low ratings or reviewers from manipulating the system. The paper should describe these mechanisms in detail, including how they would be enforced and monitored. The authors should also consider the potential for bias in author ratings and propose methods to mitigate this bias. This could involve using a combination of author ratings and other metrics to determine review quality. Thirdly, the authors should redesign the experiments to directly evaluate the effectiveness of the proposed review system. This would involve conducting experiments that measure the impact of the feedback and reward system on actual review quality and reviewer behavior. The experiments should not rely on a synthetic RQS but should instead use real-world data or a more realistic simulation of the peer review process. The authors could consider conducting a pilot study with a select group of AI conferences, collecting data on author and reviewer experiences, and comparing the performance of the proposed system with the traditional peer review process. Finally, the authors should provide more details on the implementation plan for the proposed system. This would include specifying the technologies to be used, the user interface design, and the data storage mechanisms. The paper should also address the issue of privacy, including how the identities of authors and reviewers would be protected and how the content of reviews would be kept confidential. The authors should also consider the practical challenges of implementing such a system, including the cost and the resources required. The paper should also include a discussion of the limitations of the proposed system and potential areas for future research. By addressing these weaknesses, the authors can significantly improve the quality and impact of their work.

Version Management

Version 1

Citation Tools

 **Questions**

Based on my analysis, I have several key questions that I believe are crucial for understanding and improving the proposed framework. Firstly, how exactly will the criteria for rewarding reviewers be determined, and who will be responsible for setting these standards? The paper mentions 'consistently high-quality reviews' but does not specify the metrics or the process for defining and measuring this quality. It is essential to understand how these criteria will be established and who will be responsible for ensuring their fairness and validity. Secondly, what specific mechanisms will be in place to prevent potential abuse in the feedback process? The paper mentions 'safeguarding against retaliation' but does not elaborate on the details. It is crucial to understand how the system will prevent authors from giving unfairly low ratings or reviewers from manipulating the system. Thirdly, what type of blockchain will be used for the reward system, and how will transactions be

recorded? The paper mentions 'blockchain-verified digital credentials' but does not specify the type of blockchain or the consensus mechanism. It is important to understand the technical details of the blockchain implementation and how it will ensure the integrity and transparency of the reward system. Fourthly, how will the privacy of authors and reviewers be maintained? The paper does not discuss the privacy measures that will be implemented to protect the identities of authors and reviewers or the content of the reviews. It is essential to understand how the system will ensure the confidentiality of the peer review process. Finally, what are the practical challenges of implementing such a system, and what resources will be required? The paper does not discuss the cost or the logistical challenges of implementing the proposed framework. It is important to understand the feasibility of implementing such a system in real-world academic settings. These questions target the core methodological choices and assumptions of the paper and seek clarification on critical aspects of the proposed system.

 Scores

Soundness: 1.0

Presentation: 1.0

Contribution: 1.0

Rating: 1.5

AI Review from ZGCA

 AI Review from ZGCA will be automatically processed

Version Management

Version 1

 Summary

This position paper proposes a bi-directional feedback framework for conference peer review in which authors can anonymously rate the quality of reviews, combined with a blockchain-enabled system to reward reviewers via digital credentials and a "Reviewer Impact Score." Section 3 in the paper discusses the framework (anonymous author ratings, blockchain-verified credentials). Section 4 outlines a pilot concept (author ratings, digital badges, pre/post surveys, and experiments training neural networks to predict a synthetic "Review Quality Score (RQS)," with high training/validation accuracy, and concludes that the framework improves review quality.

Citation Tools

 Strengths

- Addresses a significant and timely problem: sustaining review quality and accountability under growing submission loads.
- Proposes a bi-directional author-reviewer feedback loop intended to create accountability and incentives (Section 3).
- Calls for community discussion and reform, which is valuable given ongoing concerns about peer review.
- Outlines a pilot direction (author ratings, badges, pre/post surveys; Section 4), suggesting a path toward empirical study.
- Explicitly considers incentives for reviewers (credentials, impact score), which is often under-specified in reform proposals.

## ✖ Weaknesses

- Fundamental mismatch between the proposed social intervention and the evaluation: the paper claims to validate the framework with ML experiments on synthetic RQS (Section 5), rather than measuring human behavioral outcomes. This provides no evidence the framework improves real review quality.
- Key constructs are underspecified: no precise definition or construction of RQS or the "Reviewer Impact Score"; no rubric, annotation protocol, or reliability analysis tying scores to human judgments.
- The pilot described in Section 4 is not executed; the paper substitutes unrelated ML results (accuracy/loss on synthetic data) and overclaims causal improvement in review quality (end of Section 5).
- Novelty is limited: author-side feedback on reviews and blockchain-based reviewer incentives have been proposed previously; the paper does not clearly differentiate its mechanism or provide a new technical approach.
- Blockchain design details and necessity are not justified (on-chain/off-chain data, privacy, revocation, governance, cost, sybil resistance, collusion resistance, GDPR/CCPA compliance).
- Anti-retaliation and anonymity safeguards are asserted but not specified (e.g., secure aggregation, delayed disclosure, reviewer identity protection, moderation workflows).
- No threat model or anti-gaming mechanisms (e.g., collusive review rating rings, tit-for-tat behaviors, adversarial sybil identities) are provided.
- Clarity gaps: the paper does not describe data, measurement instruments, or how figures/metrics relate to actual review processes. It reads as a conceptual proposal with disconnected ML plots.
- Overclaiming: concluding that the framework "can improve review quality" from synthetic ML performance is not supported by the presented evidence.

## ? Questions

- How is the Review Quality Score (RQS) defined and constructed? What rubric, labels, and inter-rater reliability support it, and how does it map to review qualities valued by ACs/Senior ACs?
- What is the exact definition and formula for the "Reviewer Impact Score"? How does it aggregate ratings over time, discount noise, and protect against outliers or strategic manipulation?
- Please clarify the disconnect between Section 4 (human-subjects pilot with author ratings and surveys) and Section 5 (ML on synthetic RQS). How do human evaluations conducted? What concrete evidence links RQS predictions to improvements in actual review quality?
- What is your threat model? How do you mitigate retaliation, collusion among authors to up/down-rate reviewers, and sybil attacks? What mitigations do you detect, and how?
- Why is a blockchain necessary versus alternatives (e.g., verifiable credentials with off-chain attestations, signed logs, or existing identity systems)? What chain, costs, data stored on-chain vs off-chain, and how do you handle revocation and appeals?
- How do you ensure privacy and compliance (GDPR/CCPA) when issuing immutable credentials tied to reviewer performance? What data are public, what are hashed/committed, and how can errors be corrected?
- What governance model adjudicates disputes (e.g., contested ratings, accusations of gaming), calibrates rubrics, and updates the system over time? Who has authority and how is it accountable?
- What is your planned empirical study to measure human outcomes? For example, will you run an A/B test across areas, use AC-rated review quality rubrics, measure rebuttal usefulness, or examine decision consistency? What statistical power and pre-registration are planned?
- How will you protect author anonymity in small subcommunities where rating patterns could deanonymize identities? Do you employ secure aggregation or delayed release mechanisms?
- Can you provide ablations or simulations that model realistic human behaviors (noise, strategic play, collusion) to test robustness of the impact score and rating aggregation, prior to field deployment?
- What integration plan do you envision with existing conference management platforms (e.g., OpenReview, CMT)?
- If you keep an ML component, how will it be used in practice (e.g., outlier detection for moderation) without substituting for human evaluation?

### Version Management

Version 1

### Citation Tools

 Limitations

- Potential for gaming and collusion (e.g., coordinated rating behavior) that could distort incentives and harm reviewers or authors.
- Risk of retaliation or perceived retaliation despite anonymity, especially in small subfields; rating patterns could deanonymize participants.
- Blockchain immutability may entrench errors and create long-lived reputational harms; public attestations may leak sensitive metadata.
- Equity concerns: tokenized incentives could advantage already well-resourced reviewers, amplify prestige hierarchies, and shift focus from substantive feedback to performative metrics.
- Measurement risk: poorly calibrated or noisy author ratings could incentivize superficial positivity rather than rigorous, critical reviews.
- Operational and environmental costs of blockchain infrastructure; unclear sustainability and maintenance burden for conferences.
- Legal/ethical constraints (GDPR/CCPA) around storing performance-related data and providing rights to rectification/erasure.
- Generalizability concerns across venues, cultures, and subfields with different norms and sizes.

 Image Evaluation

Cross-Modal Consistency: 26/50

Textual Logical Soundness: 18/30

Visual Aesthetics & Clarity: 16/20

Overall Score: 60/100

Detailed Evaluation (≤500 words):

Visual ground truth (image-first)

- Figure 1/(a): RQS Training vs Validation Metric; epochs (0–200) vs metric value (0.965–1.0); blue/orange lines with cyclical dips, both  $\rightarrow \approx 1.0$ .
- Figure 1/(b): RQS Training vs Validation Loss; epochs vs loss (0–0.035); blue/orange curves with spikes at  $\approx 50/100/150$  epochs, then ne
- Figure 1 synopsis: Baseline NN shows scheduler-like resets; validation tracks training closely.
- Figure 2/(a): Training vs Validation Loss; epochs (0–150) vs loss (0–0.05); gentle plateau then sharp drop near 100 epochs; val=train.
- Figure 2/(b): Training vs Validation Metric; epochs (0–150) vs metric ( $\sim 0.95$ –1.0); rising to  $\approx 0.999$  after 110 epochs.
- Figure 2/(c): Predictions vs Ground Truth; scatter with red “line of equality”; positive correlation with variance at low GT.
- Figure 2 synopsis: “Research” setting: improved convergence and high metric; prediction scatter shows near-calibrated trend.

1. Cross-Modal Consistency

- Major 1: Figures evaluate a synthetic RQS predictor, not the proposed bi-directional/blockchain framework, so visuals cannot substantiate the central system-level claim. Evidence: “baseline...predict a synthetic Review Quality Score (RQS)” (Sec 5).
- Major 2: Experimental Setup promises a conference pilot with surveys, but no corresponding figures/tables or results are provided. Evidence: “pilot study...Surveys will be conducted pre- and post-implementation” (Sec 4).
- Minor 1: Text quotes exact accuracies (0.9981/0.9974) but figures lack numeric annotations or a table to verify. Evidence: “Results indicate high training and validation accuracy (0.9981 and 0.9974)” (Sec 5).
- Minor 2: Figure 2 caption mentions three subplots but panes lack (a,b,c) labels; readers must infer order. Evidence: “includes three subplots” (Fig. 2 caption).

2. Text Logic

- Major 1: RQS is undefined (scale, construction, dataset) and no link is made between RQS improvements and real review quality, breaking the argument from method to impact. Evidence: “synthetic Review Quality Score (RQS)” without definition (Sec 5).

Version Management

Version 1

Citation Tools

• Major 2: Claim that the framework “can improve review quality” is asserted without comparative baselines, ablations, or survey outcomes. Evidence: “analysis... confirms...improve review quality” (end of Sec 5).

• Minor 1: Method omits model architecture, loss, train/val split, and hyperparameter details necessary to reproduce. Evidence: Method/Experiments sections lack these items.

• Minor 2: Blockchain mechanism is proposed but not evaluated (no security, cost, or fairness analysis). Evidence: “blockchain-verified digital credentials” (Sec 3) with no results.

3. Figure Quality

• Major 1: No Major issues found.

• Minor 1: Missing sub-figure labels (a/b/c) in both figures reduces quick referencability. Evidence: Fig. 1 and Fig. 2 visuals show unlabeled panes.

• Minor 2: Metric y-axes start at -0.95, visually exaggerating gains; note this in captions. Evidence: Fig. 1(a), Fig. 2(b) y-range 0.965–1.0.

Key strengths:

- Clear, readable training/validation plots; sensible correlation plot.
- Timely problem statement and motivation.

Key weaknesses:

- Core claim not supported by presented evidence.
- Undefined RQS and missing experimental details.
- No results for the promised pilot or blockchain component.
- Figures lack sub-labels and numeric summaries; captions could annotate LR schedule or phase changes.

Figure-alone test: Mostly understandable; add (a/b/c) labels, numeric end-point values, and LR schedule markers.

Version Management

Version 1

 Scores

Originality: 1

Quality: 1

Clarity: 2

Significance: 2

Citation Tools

Soundness: 1

Presentation: 2

Contribution: 1

AI Review from SafeReviewer

 AI Review from SafeReviewer will be automatically processed

 Summary

This paper introduces a novel framework aimed at addressing the growing challenges within the AI conference peer review system, primarily focusing on enhancing review quality and accountability. The core contribution lies in the proposal of a bi-directional feedback mechanism, allowing authors to evaluate the reviews they receive, coupled with a blockchain-based reward system to incentivize high-quality reviewing. The authors argue that the increasing submission rates and reviewer burnout have led to a decline in review quality, and their proposed system seeks to mitigate these issues by introducing a more transparent and accountable process. The method section outlines the integration of an anonymous author rating system for reviews and the use of blockchain to issue digital badges to reviewers based on their performance, quantified by a 'Reviewer Impact Score'. The experimental section presents a baseline study using a neural network to predict a synthetic Review Quality Score (RQS) and a research study focusing on hyperparameter tuning and learning rate effects. The results indicate high training and validation accuracy for the baseline model and improvements in RQS metrics for the research study. The paper concludes by reiterating the potential of the proposed



framework and emphasizing the need for transformative reforms in peer review practices. However, the paper lacks specific details on the implementation of the bi-directional feedback mechanism, the precise calculation of the Reviewer Impact Score, and a clear connection between the experimental results and the effectiveness of the proposed framework in improving the overall peer review process. The paper's significance lies in its attempt to address a critical issue in the AI community, but its impact is limited by the lack of concrete implementation details and empirical validation.

✔ Strengths

The paper's primary strength lies in its identification of a critical and timely problem within the AI research community: the strain on the peer review process due to increasing submission rates and reviewer burnout. The authors correctly pinpoint the lack of reciprocal feedback and the absence of effective incentives for high-quality reviewing as key issues contributing to the decline in review quality. The proposed solution, a bi-directional feedback mechanism coupled with a blockchain-based reward system, is a novel and potentially transformative approach. The idea of allowing authors to evaluate the reviews they receive introduces a much-needed element of accountability into the process. Furthermore, the use of blockchain technology to issue verifiable digital credentials to reviewers based on their performance is an innovative way to incentivize high-quality contributions. The concept of a 'Reviewer Impact Score' to quantify reviewer performance, while not fully detailed, presents a promising avenue for objective evaluation. The paper also demonstrates an awareness of the need to protect authors from potential retaliation when providing feedback on reviews, which is a crucial consideration for the system's success. The authors also acknowledge the existing literature on the challenges of peer review, citing relevant works that highlight issues such as reviewer workload and biases. The paper's attempt to integrate technology, specifically blockchain, to address these challenges is a notable contribution. The paper also makes a call to action for the AI community to engage in the conversation and explore these transformative reforms, highlighting the importance of the issue and the need for collective effort to improve the peer review process.

✗ Weaknesses

The most significant weakness of this paper is the lack of detail in the methodological description of the proposed bi-directional feedback system. While the paper outlines the core components, it fails to provide specific information on how the author feedback mechanism would be implemented. For instance, the paper does not specify the criteria or metrics authors would use to evaluate reviews. Would it be a simple rating scale, or a more detailed assessment of clarity, thoroughness, and constructiveness? The paper also lacks details on the blockchain reward system. How are the digital badges earned? What specific actions or metrics contribute to a reviewer's 'Reviewer Impact Score'? How is this score calculated? The paper mentions blockchain-verified reviews but does not explain the technical implementation of this system. This lack of detail makes it difficult to assess the feasibility and practicality of the proposed system. My analysis confirms that the paper provides a high-level overview but lacks the necessary specifics for a complete understanding of the proposed framework. This is a high-confidence issue, as the absence of these details is evident throughout the method section.

Another significant weakness is the disconnect between the experimental results and the evaluation of the proposed framework. The experimental section presents a baseline study using a neural network to predict a synthetic Review Quality Score (RQS) and a research study focusing on hyperparameter tuning and learning rate effects. While the paper reports high training and validation accuracy for the baseline model and improvements in RQS metrics for the research study, it does not clearly explain how these results demonstrate the effectiveness of the bi-directional feedback and rewards framework in improving the actual peer review process. The experiments seem to focus more on the performance of a neural network predicting a synthetic score rather than directly evaluating the impact of the proposed system on review quality and reviewer behavior. The paper lacks a clear explanation of how the RQS is calculated and how it relates to the quality of the reviews. This is a high-confidence issue, as the experimental section does not provide a clear link between the reported results and the evaluation of the proposed framework's effectiveness in a real-world peer review setting. The paper also lacks a discussion of the limitations of using a synthetic RQS and the potential biases it might introduce.

Furthermore, the paper lacks a concrete plan for implementing the proposed system. The conclusion mentions future work focusing on refining the implementation and evaluating its impact, but it does not provide any specific steps or considerations for how this would be achieved. The paper does not address the practical challenges of implementing such a system, including the technical infrastructure required, the potential costs, and the need for community adoption. The paper also does not discuss the ethical considerations of using blockchain technology in this context, such as data privacy and security. This is a high-confidence issue, as the paper does not provide any details on the practical aspects of implementing the proposed framework. The paper's brevity, likely due to the constraints of a position paper, contributes to the lack of detail. The single figure, while illustrating the neural network training process, does not effectively communicate the core concepts of the proposed framework. The absence of a detailed explanation of the Review Quality Score (RQS) and its calculation further hinders understanding. The paper would benefit significantly from a more detailed elaboration of the proposed system's components and their interactions. The paper also lacks a discussion of the potential unintended consequences of the proposed system, such as reviewers focusing solely on maximizing their scores rather than providing thorough and constructive feedback. The paper also does not address the potential for gaming the system, where reviewers might collude to inflate each other's scores or authors

Version Management

Version 1

Citation Tools



might provide biased feedback based on personal disagreements with the review. These are high-confidence issues, as the paper does not address these potential problems.

 **Suggestions**

To significantly improve this paper, the authors should begin by providing a detailed and comprehensive description of the proposed bi-directional feedback mechanism. This should include specific information on the criteria and metrics that authors would use to evaluate reviews. For example, the authors could propose a multi-faceted evaluation system that includes metrics such as clarity, thoroughness, constructiveness, and adherence to ethical guidelines. They should also specify how the author feedback would be collected, stored, and used. The authors should also elaborate on the blockchain reward system, providing specific details on how the digital badges are earned, tracked, and verified. What specific actions or metrics contribute to a reviewer's 'Reviewer Impact Score'? How is this score calculated? What are the different levels of digital badges, and what do they represent? The authors should also explain the technical implementation of the blockchain system, including the choice of blockchain platform, the smart contracts used, and the security measures in place. This level of detail is crucial for assessing the feasibility and practicality of the proposed framework.

Furthermore, the experimental section needs to be revised to clearly demonstrate how the proposed system improves review quality and accountability. The authors should move beyond the use of a synthetic RQS and explore more direct measures of review quality. For example, they could conduct a pilot study where the proposed system is implemented in a real-world setting, and then compare the quality of reviews before and after implementation. They could also compare the performance of the system to traditional peer review methods. The authors should also provide a clear explanation of how the RQS is calculated and how it relates to the quality of the reviews. They should also discuss the limitations of using a synthetic RQS and the potential biases it might introduce. The experimental results should be presented in a way that directly supports the claims made about the effectiveness of the proposed framework. This could involve analyzing the correlation between reviewer scores and author feedback, or comparing the quality of reviews from reviewers with different impact scores. The authors should also consider using qualitative methods, such as surveys or interviews, to gather feedback from authors and reviewers about their experiences with the proposed system.

Finally, the paper needs to include a concrete plan for implementing the proposed system. The authors should discuss the practical challenges, including the technical infrastructure required, the potential costs, and the need for community adoption. They should also address the ethical implications of using blockchain technology in this context, such as data privacy and security. The authors should also discuss the potential unintended consequences of the system, such as reviewers focusing solely on maximizing their scores rather than providing thorough and constructive feedback. They should also address the potential for gaming the system, where reviewers might collude to inflate each other's scores or authors might provide biased feedback to receive higher scores. The authors should also consider how the system would handle disputes or appeals. The paper should conclude with a clear roadmap for future work, including specific steps for implementation and evaluation. This would make the paper more impactful and contribute to the ongoing discussion about improving the peer review process.

**Version Management**

[Version 1](#)

**Citation Tools**

 **Questions**

One of the primary questions that arises from this paper is: What specific criteria and metrics would authors use to evaluate the quality of reviews? The paper mentions an anonymous author rating system but does not provide details on the evaluation criteria. Would it be a simple rating scale, or a more detailed assessment of clarity, thoroughness, and constructiveness? How would the system ensure that author feedback is constructive and not biased or retaliatory? Another key question is: How exactly is the 'Reviewer Impact Score' calculated? The paper mentions that this score is based on the quality of feedback and consistency of participation, but it does not provide the specific formula or algorithm used. What weight is given to each factor? How is the quality of feedback assessed? Is it based on author ratings, or are there other metrics involved? A further question is: What are the potential unintended consequences of the proposed reward system? Could reviewers focus solely on maximizing their scores rather than providing thorough and constructive feedback? Could the system be gamed, for example, by reviewers colluding to inflate each other's scores? How would the system handle disputes or appeals? Additionally, I am curious about the practical aspects of implementing the blockchain system. What specific blockchain platform would be used? What are the technical requirements and costs associated with implementing and maintaining the system? How would data privacy and security be ensured? Finally, how would the authors address the potential for bias in author feedback? Could authors provide unfairly negative reviews due to personal disagreements with the review, or conversely, provide overly positive reviews due to a lack of expertise? What mechanisms would be in place to mitigate these biases and ensure that the feedback is fair and objective?

 Scores

Soundness: 1.0

Presentation: 1.0

Contribution: 1.0

Confidence: 4.25

Rating: 1.5

Keywords

Extract Keywords

Click the button to extract keywords

Insights

Extract Insights

Click the button to extract insights

Version Management

Version 1

Citation Tools